Escape speed

antocciphysics | TEACHER KEY | CPA

Learning target: Explain escape in an ideal gravitational-energy model.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 17; 3 course days later.*
Inelastic; 3 J can become deformation, thermal energy or sound. Total energy is conserved.

02. Explain a worked example (worked)
Specific K 12.5 J/kg and U-12.5 J/kg; sum zero at the threshold.

03. Look and explain (model)
No; enough initial K to rise out of the gravitational well; gravity weakens but does not abruptly stop.

04. Build the solution (practice)
10 J.

05. Use the idea (reason)
Atmosphere, thrust history, fuel mass, rotation, trajectory and engineering constraints are omitted.

06. Choose a method (mixed) *Review: Unit 4, Day 7; 13 course days later.*
+480 J; the object gains gravitational potential energy.

07. Exit check (exit)
No; it remains bound under the assumptions.

08. STRETCH: Apply it another way (stretch)
Circular speed maintains an ideal near-surface orbit; escape threshold reaches arbitrarily far without further thrust, ignoring atmosphere.

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Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 17; 3 course days later.*
Momentum 30 kg·m/s before and after. K_i=150 J; K_f=78 J. Inelastic; 72 J leaves kinetic energy.

02. Explain a worked example (worked)
Specific K 12.5 J/kg and U-12.5 J/kg; sum zero at the threshold.

03. Represent and interpret (model)
No; enough initial K to rise out of the gravitational well; gravity weakens but does not abruptly stop.

04. Solve and justify (practice)
Circular√50=7.07 m/s; escape√100=10 m/s; ratio√2.

05. Analyze the reasoning (reason)
Atmosphere, thrust history, fuel mass, rotation, trajectory and engineering constraints are omitted.

06. Choose a method (mixed) *Review: Unit 4, Day 7; 13 course days later.*
ΔU=-480 J; gravity does +480 J work.

07. Exit check (exit)
New escape speed is half.

08. STRETCH: Apply it another way (stretch)
Escape speed scales√(M/R); first√2 times, second unchanged.